AI-Powered Next-Generation Personal Accident Insurance Platform

(A Scalable, Automated, Trust-First Digital Insurance Ecosystem)

ABSTRACT
The Complete Accident Insurance system is a structured, rule-based, AI-assisted insurance management platform developed using .NET 8 and SQL Server. The system manages three insurance products: Individual Personal Accident Plan (18–65 years), Family Accident Plan (18–65 based on eldest member), and OverAge Accident Plan (66–80 years). It integrates slab-based pricing, centralized risk multiplier configuration, automated premium calculation, AI-powered Lead Generation Score (LGS), load-balanced agent assignment, structured claim management, and commission tracking. The objective is to build a beginner-friendly yet professionally structured backend that combines insurance domain logic with AI-assisted engagement while maintaining clean architecture, naming conventions, and centralized exception handling.

PROBLEM STATEMENT
Traditional accident insurance processes rely on manual underwriting, unstructured lead management, uneven agent distribution, and delayed claim processing. There is no structured engagement tracking before purchase, and many systems lack automated premium calculation using standardized risk factors. Additionally, most insurance platforms do not provide OverAge accident coverage beyond 65 years. The problem is to build a simplified yet intelligent digital insurance platform that automates underwriting rules, ensures fair agent workload distribution, captures AI-based lead intent, and processes claims systematically while remaining understandable for beginner developers.

DATABASE DESIGN OVERVIEW

Database Name:
CompleteAccidentInsurance

Naming Convention Style:

- Tables → PascalCase (singular noun)
- Columns → PascalCase
- Primary Key → Id
- Foreign Key → EntityNameId
- Date Fields → CreatedAt, UpdatedAt
- Status Fields → Status
- Boolean Fields → IsActive

SECTION 1: ADMIN

Table: AdminUser

Column Name	Type	Description
Id	UNIQUEIDENTIFIER (PK)	Primary Key
FullName	NVARCHAR(150)	Admin Name
OfficialEmail	NVARCHAR(150)	Login Email
PasswordHash	NVARCHAR(500)	Encrypted Password
IsActive	BIT	Account Status
CreatedAt	DATETIME	Creation Date
UpdatedAt	DATETIME	Last Modified

Admin login requires OfficialEmail and Password.

Admin responsibilities include product creation, risk configuration, agent creation, monitoring analytics, and system governance.

SECTION 2: AGENT

Table: InsuranceAgent

Column Name	Type
Id	UNIQUEIDENTIFIER (PK)
AgentCode	NVARCHAR(50)
FullName	NVARCHAR(150)
OfficialEmail	NVARCHAR(150)
MobileNumber	NVARCHAR(20)
AadhaarNumber	NVARCHAR(20)
PanNumber	NVARCHAR(20)
ResidentialAddress	NVARCHAR(300)
DateOfBirth	DATE
CommissionPercentage	DECIMAL(5,2)
TemporaryPasswordHash	NVARCHAR(500)

Column Name	Type
IsPasswordResetRequired	BIT
IsActive	BIT
CreatedByAdminId	UNIQUEIDENTIFIER (FK)
CreatedAt	DATETIME
UpdatedAt	DATETIME

Agent login requires OfficialEmail and Password.

Agents receive leads automatically, review applications, approve/reject policies, assist customers, and earn commission.

SECTION 3: CLAIMS OFFICER

Table: ClaimsOfficer

Column Name	Type
Id	UNIQUEIDENTIFIER (PK)
OfficerCode	NVARCHAR(50)
FullName	NVARCHAR(150)
OfficialEmail	NVARCHAR(150)
MobileNumber	NVARCHAR(20)
AadhaarNumber	NVARCHAR(20)
PanNumber	NVARCHAR(20)
Department	NVARCHAR(100)
TemporaryPasswordHash	NVARCHAR(500)
IsPasswordResetRequired	BIT
IsActive	BIT
CreatedByAdminId	UNIQUEIDENTIFIER (FK)
CreatedAt	DATETIME
UpdatedAt	DATETIME

Claims Officer login requires OfficialEmail and Password.

Claims Officers validate policy status, verify documents, and approve or reject claims.

SECTION 4: CUSTOMER

Table: Customer

Column Name	Type
Id	UNIQUEIDENTIFIER (PK)
CustomerCode	NVARCHAR(50)
FullName	NVARCHAR(150)
EmailAddress	NVARCHAR(150)
MobileNumber	NVARCHAR(20)
DateOfBirth	DATE
Gender	NVARCHAR(20)
AadhaarNumber	NVARCHAR(20)
PanNumber	NVARCHAR(20)
ResidentialAddress	NVARCHAR(300)
PasswordHash	NVARCHAR(500)
IsEmailVerified	BIT
IsActive	BIT
CreatedAt	DATETIME
UpdatedAt	DATETIME

Customer registers with FullName, EmailAddress, MobileNumber, DateOfBirth, Password, ConfirmPassword.

Customer login requires EmailAddress and Password.

SECTION 5: INSURANCE PRODUCT MASTER

Table: InsuranceProduct

Column Name	Type
Id	UNIQUEIDENTIFIER (PK)
ProductName	NVARCHAR(150)
ProductCategory	NVARCHAR(50)
MinimumEligibleAge	INT
MaximumEligibleAge	INT
PolicyValidityInMonths	INT
Description	NVARCHAR(MAX)
KeyBenefits	NVARCHAR(MAX)
WaitingPeriodInDays	INT
IsActive	BIT
CreatedAt	DATETIME
UpdatedAt	DATETIME

Products include Individual, Family, and OverAge.

SECTION 6: COVERAGE PLAN

Table: CoveragePlan

Column Name	Type
Id	UNIQUEIDENTIFIER (PK)
InsuranceProductId	UNIQUEIDENTIFIER (FK)
CoverageAmount	DECIMAL(18,2)
BasePremium	DECIMAL(18,2)
PremiumFrequencyAllowed	NVARCHAR(50)
IsActive	BIT
CreatedAt	DATETIME

CoveragePlan defines slab-based pricing.

SECTION 7: RISK MASTER CONFIGURATION

Table: AgeBandRiskMultiplier

Column	Type
Id	UNIQUEIDENTIFIER
MinimumAge	INT
MaximumAge	INT
MultiplierValue	DECIMAL(5,2)
ApplicableProductCategory	NVARCHAR(50)
IsActive	BIT

Table: OccupationRiskCategory

Column	Type
Id	UNIQUEIDENTIFIER
OccupationName	NVARCHAR(150)
RiskLevel	NVARCHAR(50)
MultiplierValue	DECIMAL(5,2)
IsActive	BIT

Table: LifestyleRiskFactor

Column	Type
Id	UNIQUEIDENTIFIER
FactorName	NVARCHAR(100)
MultiplierValue	DECIMAL(5,2)
IsActive	BIT

Premium formula:
FinalPremium = BasePremium × AgeMultiplier × OccupationMultiplier × LifestyleMultiplier

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SECTION 8: POLICY APPLICATION

Table: PolicyApplication

Column	Type
Id	UNIQUEIDENTIFIER
CustomerId	UNIQUEIDENTIFIER (FK)
CoveragePlanId	UNIQUEIDENTIFIER (FK)
SelectedOccupationId	UNIQUEIDENTIFIER (FK)
IsSmoker	BIT
IsDrinker	BIT
TravelFrequency	NVARCHAR(100)
CalculatedPremiumAmount	DECIMAL(18,2)
ApplicationStatus	NVARCHAR(50)
AssignedAgentId	UNIQUEIDENTIFIER (FK)
CreatedAt	DATETIME
UpdatedAt	DATETIME

SECTION 9: CUSTOMER DOCUMENTS

Table: CustomerDocument

Column	Type
Id	UNIQUEIDENTIFIER
PolicyApplicationId	UNIQUEIDENTIFIER (FK)
DocumentType	NVARCHAR(100)
FileUrl	NVARCHAR(500)
UploadedAt	DATETIME

SECTION 10: PAYMENT

Table: PremiumPayment

Column	Type
Id	UNIQUEIDENTIFIER
PolicyApplicationId	UNIQUEIDENTIFIER
PaymentReferenceNumber	NVARCHAR(100)
PaymentAmount	DECIMAL(18,2)
PaymentFrequency	NVARCHAR(50)
PaymentStatus	NVARCHAR(50)
PaidOn	DATETIME

SECTION 11: ACTIVE POLICY CONTRACT

Table: ActivePolicyContract

Column	Type
Id	UNIQUEIDENTIFIER
PolicyApplicationId	UNIQUEIDENTIFIER
PolicyNumber	NVARCHAR(100)
PolicyStartDate	DATE
PolicyEndDate	DATE
CurrentStatus	NVARCHAR(50)
CreatedAt	DATETIME

SECTION 12: CLAIM MANAGEMENT

Table: InsuranceClaim

Column	Type
Id	UNIQUEIDENTIFIER
ActivePolicyContractId	UNIQUEIDENTIFIER
ClaimReferenceNumber	NVARCHAR(100)

Column	Type
IncidentDate	DATE
ClaimDescription	NVARCHAR(MAX)
ClaimAmountRequested	DECIMAL(18,2)
ClaimStatus	NVARCHAR(50)
ReviewedByOfficerId	UNIQUEIDENTIFIER
DecisionRemarks	NVARCHAR(MAX)
CreatedAt	DATETIME

SECTION 13: AI LEAD GENERATION

Table: Lead

Column	Type
Id	UNIQUEIDENTIFIER
CustomerId	UNIQUEIDENTIFIER
InterestedProductId	UNIQUEIDENTIFIER
CurrentLeadScore	DECIMAL(5,2)
LeadStatus	NVARCHAR(50)
AssignedAgentId	UNIQUEIDENTIFIER
CreatedAt	DATETIME
UpdatedAt	DATETIME

Table: ChatMessage

Column	Type
Id	UNIQUEIDENTIFIER
LeadId	UNIQUEIDENTIFIER
SenderType	NVARCHAR(50)
SenderId	UNIQUEIDENTIFIER
MessageContent	NVARCHAR(MAX)

Column	Type
SentAt	DATETIME

Lead Generation Score increases based on engagement and buying intent. When threshold is reached, system auto-assigns an agent using load balancing.

REQUIREMENTS FOR FRONTEND AND BACKEND DEVELOPMENT

The Complete Accident Insurance system requires a full-stack implementation using a structured backend (ASP.NET Core .NET 8 + SQL Server) and a modern frontend (Angular 20/21 standalone architecture or equivalent SPA framework). The backend must expose REST APIs with JWT authentication and role-based authorization. The frontend must provide role-specific dashboards for Admin, Agent, Claims Officer, and Customer. The system must include policy management, risk configuration, premium calculation, AI-based lead engagement, agent assignment, payment flow, and claims processing. A global exception handler must be implemented in the backend. The system should follow clean naming conventions, simple logic, and beginner-friendly structured coding practices.

SYSTEM ARCHITECTURE

The system follows a layered architecture:

1. Presentation Layer (Frontend – Angular SPA)
    - Role-based dashboards
    - Policy browsing interface
    - Chat interface (AI + Agent)
    - Application and claims forms
    - Admin management panels
2. API Layer (ASP.NET Core .NET 8)
    - REST Controllers
    - JWT Authentication
    - Role-based Authorization
    - Global Exception Middleware
3. Business Logic Layer
    - Premium Calculator (Multiplier-based logic)
    - Lead Score Calculator (Rule-based scoring)
    - Agent Load Balancer
    - Commission Calculator

4. Data Access Layer
    - Entity Framework Core
    - SQL Server Database (CompleteAccidentInsurance)
    - Migrations for schema control
5. AI Engagement Layer
    - Chat interaction handler
    - Intent extraction
    - Lead Generation Score update
    - Escalation trigger to agent

The architecture ensures separation of concerns while keeping the implementation simple and readable.

SYSTEM ROLES

1. Admin
    - Creates insurance products and coverage plans
    - Configures risk multipliers (age, occupation, lifestyle)
    - Creates agents and claims officers
    - Monitors analytics, revenue, workload, and performance
    - Controls activation and deactivation of system modules
2. Customer
    - Registers and logs in
    - Browses products and coverage plans
    - Interacts with AI chatbot
    - Applies for policy
    - Makes payments
    - Raises claims
3. Agent
    - Receives assigned leads and applications
    - Reviews and approves/rejects applications
    - Communicates with customers via chat
    - Follows up on pending payments

    - Earns commission based on successful conversions
4. Claims Officer
    - Reviews submitted claims
    - Verifies documents and eligibility
    - Approves or rejects claims
    - Maintains claim audit records

END-TO-END WORKFLOW SUMMARY

The system workflow begins when a customer registers and logs in. After authentication, the customer explores available insurance products (Individual, Family, OverAge). When a customer shows interest in a product, a Lead is created and AI chatbot interaction begins. The chatbot gathers structured information and calculates a Lead Generation Score (LGS). If the score crosses a defined threshold, the system assigns the least-loaded agent automatically and initiates live chat between the agent and the customer.

If the customer proceeds directly to purchase, they fill required risk details such as age, occupation, and lifestyle factors. The system calculates premium automatically using predefined multipliers. A Policy Application record is created and assigned to an agent through load-balancing logic. The agent reviews the application and either approves or rejects it.

If approved, the customer selects payment frequency and completes payment. After successful payment, an ActivePolicyContract is generated with policy number and validity dates. The agent remains associated with the policy for follow-up and support.

In case of an accident, the customer can submit a claim from their dashboard by uploading required documents. A Claims Officer reviews the claim, verifies policy validity and eligibility, and approves or rejects it. The decision is recorded in the system for audit tracking.

Throughout the process, Admin monitors analytics including total policies sold, premium revenue, claim ratio, lead conversion rate, and agent performance.

WORKFLOW SUMMARY FROM LOGIN TO PAYMENT

1. Customer registers using email and password.
2. Customer logs in via JWT authentication.
3. Customer browses insurance products.
4. Customer clicks Interested → AI chat begins → Lead Score increases.
5. If Lead Score threshold reached → Agent assigned automatically.
6. Customer selects coverage plan and enters risk details.
7. System calculates premium using multiplier-based formula.
8. PolicyApplication is created.

9. Agent reviews and approves application.
10. Customer selects payment frequency and completes payment.
11. Active policy contract is generated.
12. Policy lifecycle begins under assigned agent supervision.

This workflow ensures automation, intelligent engagement, and structured underwriting while maintaining simplicity and clarity in implementation.

USP

1. AI-driven Lead Generation Score.
2. OverAge accident insurance product (66–80).
3. Slab-based premium calculation engine.
4. Automated load-based agent assignment.
5. Centralized global exception handling middleware.

CONCLUSION

The Complete Accident Insurance platform integrates structured insurance logic with AI-assisted engagement and automated workflow management. It provides slab-based pricing, multiplier-driven underwriting, intelligent lead conversion, fair agent distribution, structured claims management, and scalable backend architecture. The system remains beginner-friendly in implementation while maintaining professional standards in database design, naming conventions, and centralized error handling.